Remark 5.26.2. Statement (ii) of Theorem 5.26.1 is equivalent to the same statement with $\mathbb{Q}$-span replaced by $\mathbb{C}$-span. Indeed, consider the matrix whose columns consist of the coefficients of the decomposition of $\mathrm{Ind}_H^G V$ (for various H, V) with respect to the irreducible representations of G. Then both statements are equivalent to the condition that the rows of this matrix are linearly independent.

Proof. *Proof that (ii) implies (i).* Assume that $g \in G$ does not belong to any of the subgroups $H \in X$. Then, since X is conjugation invariant, it cannot be conjugated into such a subgroup. Hence by the Frobenius formula, $\chi_{\mathrm{Ind}_H^G(V)}(g) = 0$ for all $H \in X$ and V. So by (ii), for any irreducible representation W of G, $\chi_W(g) = 0$. But irreducible characters span the space of class functions, so any class function vanishes on g, which is a contradiction.

Proof that (i) implies (ii). Let U be a virtual representation of G over $\mathbb{C}$ (i.e., a linear combination of irreducible representations with nonzero integer coefficients) such that $(\chi_U, \chi_{\mathrm{Ind}_H^G V}) = 0$ for all H, V. So by Frobenius reciprocity, $(\chi_{U|_H}, \chi_V) = 0$. This means that χ_U vanishes on H for any $H \in X$. Hence by (i), χ_U is identically zero. This implies (ii) (because of Remark 5.26.2). $\square$

Corollary 5.26.3. *Any irreducible character of a finite group is a rational linear combination of induced characters from its cyclic subgroups.*

5.27. Representations of semidirect products

Let G, A be groups and let $\phi : G \to \mathrm{Aut}(A)$ be a homomorphism. For $a \in A$, denote $\phi(g)a$ by $g(a)$. The **semidirect product** $G \ltimes A$ is defined to be the product $A \times G$ with multiplication law

$$(a_1, g_1)(a_2, g_2) = (a_1 g_1(a_2), g_1 g_2).$$

Clearly, G and A are subgroups of $G \ltimes A$ in a natural way.

We would like to study irreducible complex representations of $G \ltimes A$. For simplicity, let us do it when A is abelian.

In this case, irreducible representations of A are 1-dimensional and form the character group $A^\vee$, which carries an action of G. Let O be an orbit of this action, $x \in O$ a chosen element, and G_x the